Course Code	**DESIGN OF EARTHQUAKE RESISTANT STRUCTURES** **Professional Elective-II**	L	T	P	C
23A01604a		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental concepts of engineering seismology, including earthquake phenomena, seismic waves, and measuring instruments.
2. **Analyze** the principles of structural vibrations, degrees of freedom, and dynamic response of structures to earthquake ground motions.
3. **Evaluate** conceptual design strategies, seismic design principles, and methods for improving earthquake resistancein structures.
4. **Apply** earthquake-resistant design principles to reinforced concrete and masonry buildings using IS codes and lateral force methods.
5. **Assess** the role of structural walls, non-structural elements, and ductility considerationsin enhancing earthquake resistance.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. **Explain** earthquake mechanisms, seismic waves, and seismic zones, including measuring techniques and instruments.
2. **Analyze** vibratory systems, single-degree-of-freedom (SDOF) models, damping effects, and earthquake-induced dynamic forces.
3. **Evaluate** conceptual design strategies, ductility factors, and seismic design methods for ensuring structural resilience.
4. **Apply**IS code provisions and lateral force methods for seismic design of reinforced concrete and masonry buildings.
5. **Assess** the significance of structural walls, non-structural elements, and ductile detailingin enhancing earthquake resistance.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	-	-	1	-	3	1	-	2	-	2	2	2
CO -2	3	-	-	-	2	-	3	1	-	2	-	-	-	2
CO -3	3	-	-	3	-	-	3	2	-	2	-	-	-	3
CO -4	3	2	-	3	-	-	3	2	-	2	2	2	3	3
CO -5	3	3	-	3	-	-	3	2	-	2	2	2	3	3

UNIT – I

Engineering Seismology: Earthquake Phenomenon - Cause of Earthquakes-Faults- Plate Tectonics- Seismic Waves- Terms Associated With Earthquakes-Magnitude/Intensity of An Earthquake-Scales- Energy Released-Earthquake Measuring Instruments Seismogram - Seismoscope, Seismograph, - Strong Ground Motions- Seismic Zones of India.

Theory of Vibrations: Elements of A Vibratory System- Degrees of Freedom-Continuous System- Lumped Mass Idealization-Oscillatory Motion-Simple Harmonic Motion-Free Vibration of Single Degree of Freedom (SDOF) System- Undamped and Damped-Critical Damping-Logarithmic Decrement-Forced Vibrations-Harmonic Excitation-Dynamic Magnification Factor-Excitation By Rigid Based Translation for SDOF System-Earthquake Ground Motion.

UNIT – II

Conceptual Design: Introduction-Functional Planning-Continuous Load Path-Overall form-Simplicity and Symmetry-Elongated Shapes-Stiffness and Strength-Horizontal and Vertical Members-Twisting of Buildings-Ductility-Ductility Relationships-Flexible Buildings-Framing Systems-Choice of Construction Materials-Unconfined Concrete-Confined Concrete-Masonry-Reinforcing Steel.

Introduction to Earthquake Resistant Design: Seismic Design Requirements-Regular and Irregular Configurations-Basic Assumptions-Design Earthquake Loads-Basic Load Combinations-Permissible Stresses-Seismic Methods of Analysis-Factorsin Seismic Analysis-Equivalent Lateral force Method.

UNIT – III

Reinforced Concrete Buildings: Principles of Earthquake Resistant Deign of RC Members-Structural Models for Frame Buildings - Seismic Methods of Analysis- Is Code Based Methods for Seismic Design - Vertical Irregularities - Plan Configuration Problems- Lateral Load Resisting Systems- Determination of Design Lateral forces as Per Is 1893 (Part-1):2016- Equivalent Lateral force Procedure- Lateral Distribution of Base Shear.

UNIT – IV

Masonry Buildings: Introduction- Elastic Properties of Masonry Assemblage- Categories of Masonry Buildings- Behaviour of Unreinforced and Reinforced Masonry Walls- Behavior of Walls- Box Action and Bands- Behaviour of Infill Walls- Improving Seismic Behaviour of Masonry Buildings- Load Combinations and Permissible Stresses- Seismic Design Requirements- Lateral Load Analysis of Masonry Buildings.

UNIT – V

Structural Walls and Non-Structural Elements: Strategies in The Location of Structural Walls- Sectional Shapes- Variations in Elevation- Cantilever Walls Without Openings – Failure Mechanism of Non-Structures- Effects of Non-Structural Elements On Structural System- Analysis of Non-Structural Elements- Prevention of Non-Structural Damage Ductility Considerations in Earthquake Resistant Design of RC Buildings: Introduction-Impact of Ductility- Requirements for Ductility- Assessment of Ductility- Factors Affecting Ductility- Ductile Detailing Considerations as Per Is 13920-2016 - Behaviour of Beams, Columns and Joints in RC Buildings During Earthquakes

TEXT BOOKS:

1. Earthquake Resistant Design of structures – S. K. Duggal, Oxford University Press
2. Earthquake Resistant Design of structures – PankajAgarwal and Manish Shrikhande, PrenticeHall of India Pvt. Ltd.

REFRENCE BOOKS:

1. Seismic Design of Reinforced Concrete and Masonry Building – T. Paulay and M.J.N. Priestly,John Wiley & Sons.
2. Eartquake Resistant Design ofBuidling structures by VinodHosur, Wiley India Pvt. Ltd.
3. Elements of Mechanical Vibration by R.N.Iyengar, I.K.International Publishing House Pvt. Ltd.
4. Masory and Timber structures including earthquake Resistant Design –AnandS.Arya, Nemchand& Bros
5. Earthquake Tips – Learning Earthquake Design and Construction, C.V.R. Murthy
6. BIS Codes: 1. IS 1893(Part-1):2016 or Latest codes;. 2. IS 13920:2016. 3. IS 4326. 4. IS 456:2000 or latest.

Online Learning Resources:

https://nptel.ac.in/courses/105107204